

4.1.2 Results for $\mathcal{D}_p$

For varying p , we train two-layer transformer models with embedding dimensions of 128 on $\mathcal{D}_p$ using both ordinary and myopic gradient descent. Full architecture and training details are provided in Section D.1.

Examining the performance of each linear probe against $\sin(bx_{n-i})$ for varying i , we find strong evidence that the transformer model with vanilla training is indeed pre-caching $\sin(bx_n)$, possibly in order to implement Algorithm 1. Indeed, in Figure 2,

- The zeroth hidden state (i.e., the sum of the input and position embeddings) at position n is correlated with only x_n .
- The first hidden state is correlated with $\sin(bx_n)$ but not correlated with any $\sin(bx_{n-i})$ for $i > 0$.
- The second hidden state (immediately before the output unembedding) is correlated with $\sin(bx_{n-i})$ for each $0 \leq i \leq a$.

Further, looking at the per-neuron correlations in Figure 3, we see that $\sin(bx_{n-i})$ for $1 \leq i \leq a$ are all correlated with a single 1-d subspace of the second hidden state (they share the same striping pattern); this is the subspace storing $\sum_{i=1}^a \sin(bx_{n-i})$. Meanwhile, $\sin(bx_n)$, as well as many of the x_{n-i} , are located in various other 1-d subspace of the second hidden state; these terms are all left over in the residual stream from previous layers, and are cleaned up only by the output unembedding.

On the other hand, in Table 1, the myopic models perform significantly worse. The per-neuron correlations in Figure 3 suggest that the myopic model may be implementing a crude approximation of Algorithm 2. This suggests that the synthetic setting has an inherently high *myopia gap*—it is impossible for the transformer model to do well without pre-caching.

p	Vanilla	Myopic
0.01	0.096	1.10
0.1	0.016	0.97
0.3	0.0030	1.03
1.0	0.0074	1.26

Table 1: Normalized Huber loss $\mathcal{L}/p$ for vanilla and myopic models trained and evaluated on $\mathcal{D}_p$ for each p in our synthetic setting. For reference, the trivial model that always outputs zero attains a Huber loss of 1.26.

4.2 Multiplication

In addition to the above $\mathcal{D}_p$ synthetic data setting, we also measure the myopia gap on the task of natural number multiplication. In particular, we find evidence suggesting that pre-caching is responsible for model computation on filler tokens, in the sense of Pfau et al. (2024). See Appendix D.3 for details.

5 Natural language experiments

5.1 GPT-2’s myopia gap

In order to measure the extent to which transformer models learn to pre-cache on natural language data, we estimate both the myopia gap (Definition 3) in this setting as well as the local myopia bonus (Definition 4) of a transformer model with vanilla pre-training. Experiments in this subsection use the 124M-parameter GPT-2 architecture; see Table 4 in Appendix D for configuration details.

We train all models (vanilla and myopic) from random initialization for one epoch on 4.6M sequences from the MS MARCO dataset (Nguyen et al., 2016), truncated to length 64. To estimate the local myopia bonus of the vanilla model, we train another model from random initialization with the same architecture, but with past hidden states sourced from the frozen

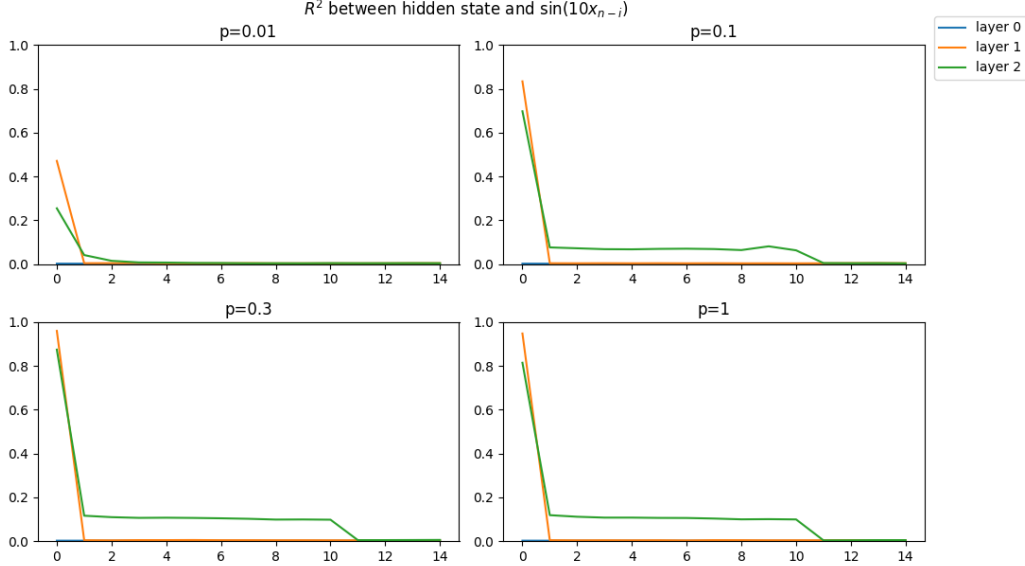


Figure 2: Empirical R^2 between linear probes fit on each layer of vanilla transformer models trained on $\mathcal{D}_p$ for $p \in \{0.01, 0.1, 0.3, 1\}$ to targets $\sin(bx_{n-i})$. Computed over 50 000 samples from $\mathcal{D}_1$.

vanilla model during both training and evaluation.⁷ See Appendix B for implementation details.

As baseline, we also train a “transformer bigram” model, a model with an identical architecture but all off-diagonal key/value states zeroed out.

5.1.1 GPT-2 results

From Table 2, the estimated myopia gap in this setting is $3.40 - 3.28 = \mathbf{0.12}$ cross entropy, while the local myopia bonus of the vanilla model is $3.28 - 3.26 = \mathbf{0.02}$.

The nonzero myopia gap suggests that pre-caching may provide a small positive benefit. Indeed, in Figure 4, we see that the myopic model outperforms the vanilla model at the beginning of the sequence, since it can allocate all compute to next-token prediction, but quickly falls behind as the length of the past increases, since it suffers from a lack of pre-cached information from earlier forward passes.⁸

However, note that this gap is much smaller than that between the vanilla model and the transformer bigram model (Table 2). That is, the myopic model is still able to leverage past information (breadcrumbs) to a significant extent, even if they optimized only for the present inference task. That the local myopia gap is near zero further supports this direction—the model learned through vanilla training does not trade off significantly between features useful for the present and pre-caching for the future.

Model	Cross-entropy
Vanilla	3.28
Myopic	3.40
Local myopic	3.26
Transformer bigram	5.33

Table 2: Validation cross-entropy loss obtained by GPT-2 with vanilla and myopic training

⁷Note that this “local myopic” model attains slightly *better* performance than the vanilla model; each forward pass can focus purely on next-token prediction, since past hidden states are supplied by a separate model.

⁸We use a sliding window over PG-19 (Rae et al., 2019) samples, which comprise longer sequences, in order to reduce noise in our per-position loss estimates. The distribution shift between MS MARCO and PG-19 does not affect the relative comparison of myopic and vanilla GPT-2.

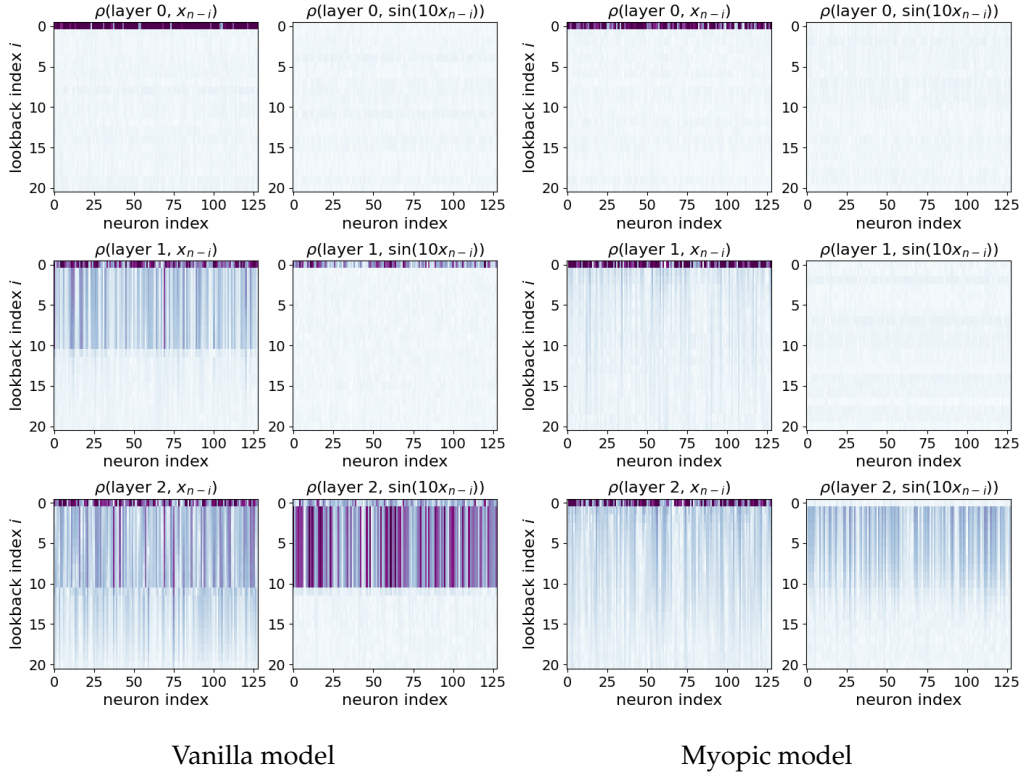


Figure 3: Empirical correlations between each hidden state neuron and x_{n-i} or $\sin(bx_{n-i})$. Models are vanilla (left two columns) and myopic (right two columns) transformers trained on $\mathcal{D}_{0.3}$.

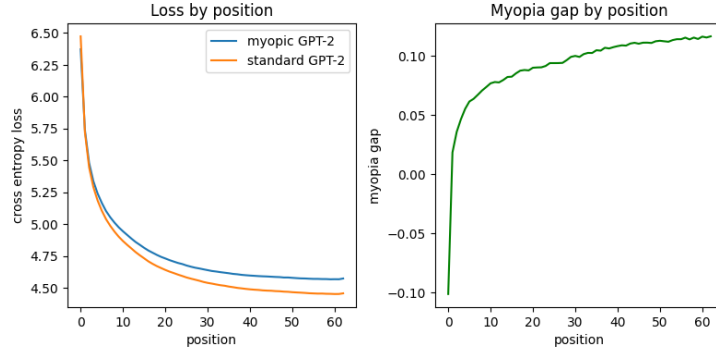


Figure 4: Cross-entropy loss of vanilla and myopic GPT-2 models by token position, and their difference. Evaluated on a sliding window over a 100K-token sample text from the PG-19 dataset (Rae et al., 2019). Aggregate cross-entropy losses on this sample are 4.67 (vanilla) and 4.77 (myopic).

5.2 Myopia gap scaling

One might suppose that the relatively small myopia gap of GPT-2 is due to the relative simplicity of the small architecture we consider, and that larger language models might exhibit a more pronounced myopia gap.

To test this, we train both vanilla and myopic transformers from the Pythia LLM suite (Biderman et al., 2023), ranging in size from 14M to 2.8B parameters, on one epoch of 10M